RFID LIBRARY AUTOMATION



Network Toll Solution

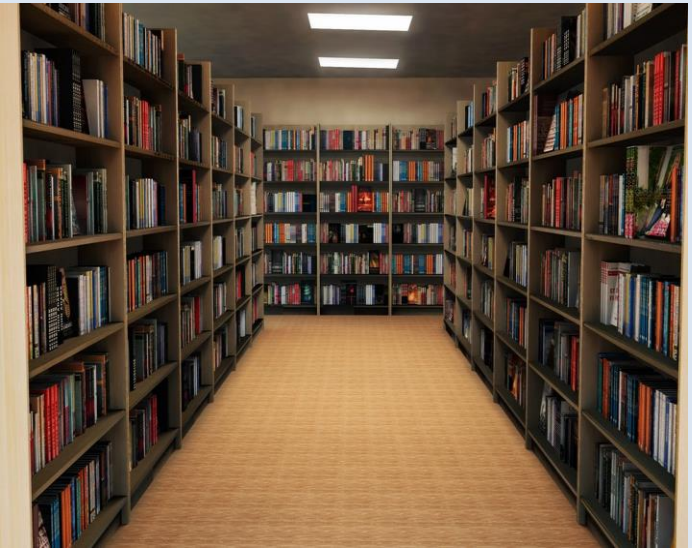
www.networktoll.com
Unit No. 1008, Tower 3, Aastha Greens, Sector 4, Greater Noida West, U.P. 201306





PRESENTING

“NTS LIB”







1. Company Profile
2. Why us?
3. Traditional Indian Library
4. Issues and Challenges
5. Solution Benefits
6. RFID VS Barcode
7. Components
8. Clients

COMPANY PROFILE

- Incepted with an objective to make “tracking” time and cost effective.
- Core domain AIDC vertical with focus on RFID technology
- State of art infrastructure with Internal R&D and manufacturing unit in Delhi and Gurgaon
- Lead by a team of IIT’ians .
- Team strength of 20 plus. Experience of more than a decade.
- Global experience in automating 30 + libraries
- First company to automate Delhi University Library.
- Experience of automating one of the largest libraries in India - RGPV
- Among top 10 RFID automation service providers in India. CIO 2016
- Experience of integration with KOHA, TCS, LIBERTY and other leading softwares in the market. Can be integrated with SLIM and E gran
- ISO 9001:2008 certification . Full compliance to NCIP Protocol for Libraries.



Network Toll Solution

- Experience of more than a decade in RFID Library Automation.
- Strong project management skills.
- Quick turn around time. 24*7 technical call centre help
- Integration experience with almost all possible ILMS in market.
- Customized hardware design to match library theme.
- In house software development, installation & commissioning, R&D team.

- High quality products with comprehensive AMC facility.
- Three way approach of working

SYSTEM INTEGRATOR	SOLUTION ARCHITECHT	DATA ANALYST

- Focus on ROI
- Competitive pricing.
- Facility for data entry / tagging etc

Traditional Indian Libraries

- Still using barcode technology.
- Generally a guard is deputed at exit point to check book theft.
- Inventory is counted manually.
- Library users / patrons are given paper cards for transaction of books.
- On purchase, book information is manually entered in registers.
- Working hours : 9 AM – 7 PM
- Library staff is generally engaged in circulation or data management work.
- Finding a particular book is a tedious work.
- Proprietorship ILMS.

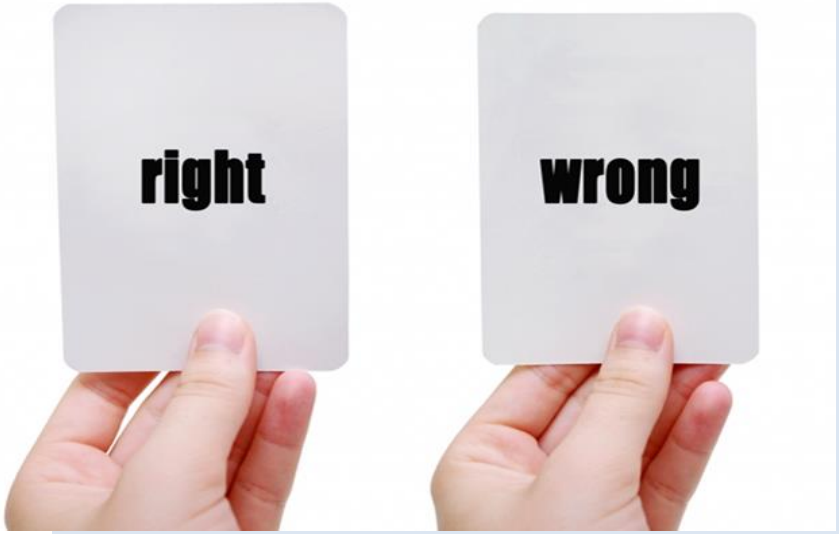




- Theft. Loss of books, journals, CD's, project reports etc.
- Poor inventory accuracy.
- Long queue at circulation desk.
- Lack of valuable information to patrons
- Inefficient use of trained staff.
- Miss – shelved books.
- Time constraint.
- Increased material handling cost.







How can a librarian spend more time educating the students when they spend most of their day coping with the other issues of the library.

+ We've got the

solution

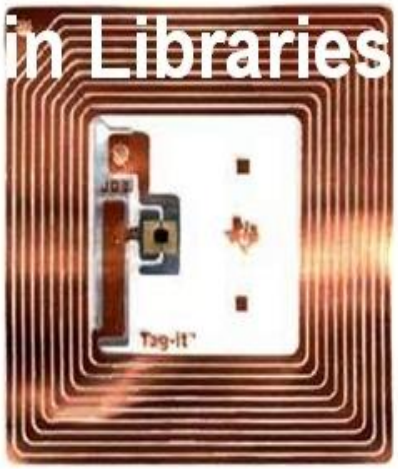


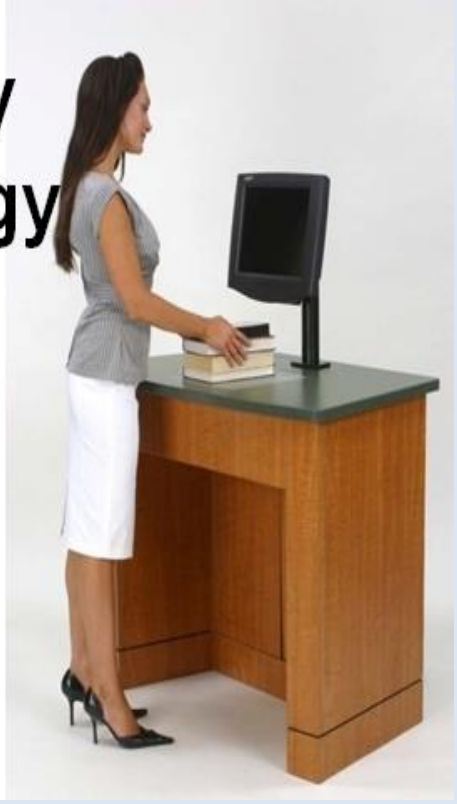
Library Management with RFID Technology



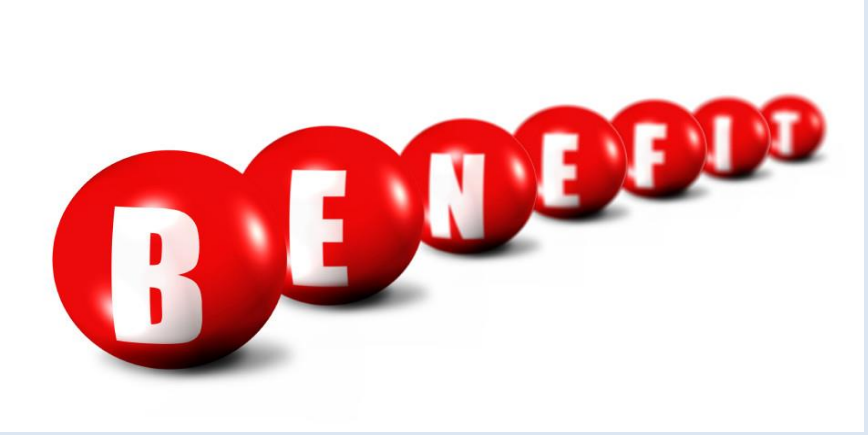


RFID (Radio Frequency IDentification) technology in Libraries





HIGH FREQUENCY RFID (HF)



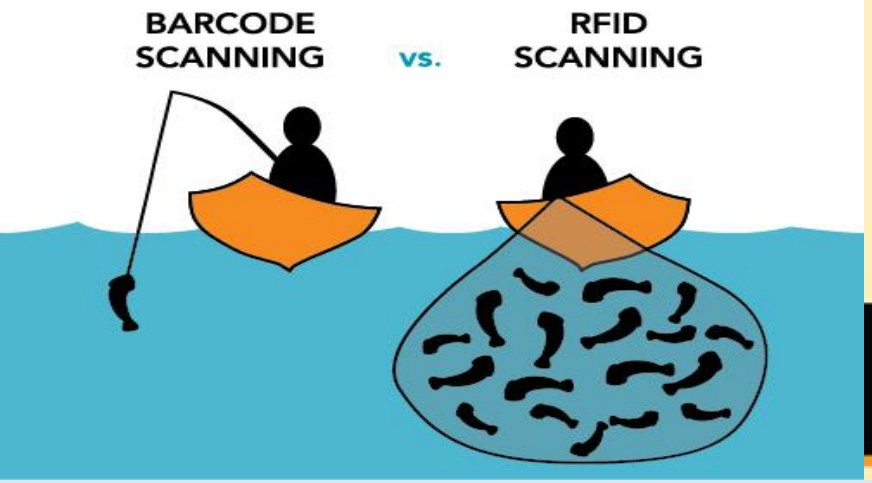


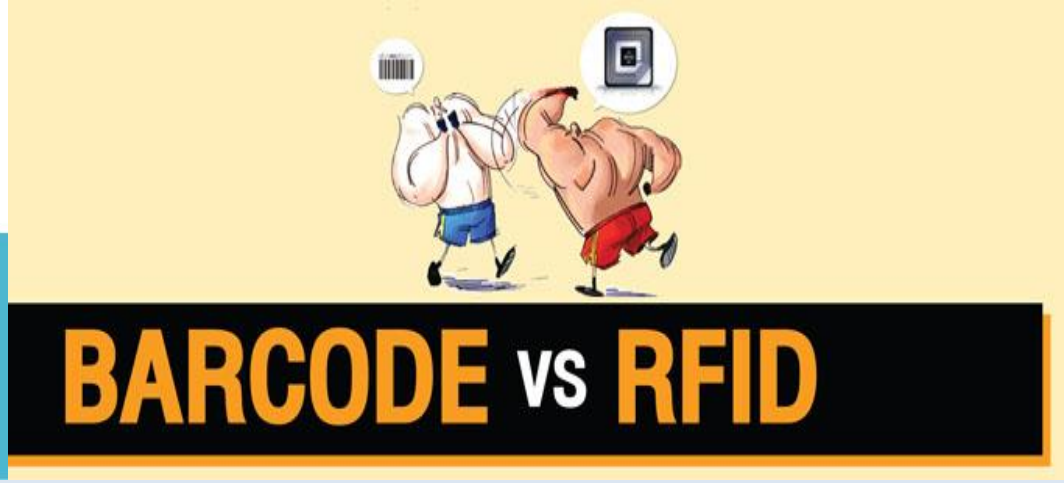
For Library Staff

- Saves time in inventory and stock verification.
- Quick circulation.
- Prevents theft.
- Better utilization of resources.
- Cost effective.
- 24*7 library.
- Helps in locating mis shelved books.
- Paper less transaction.
- Better and smooth back end operations.
- Branding by Updation to new technology.
- Customer / patron satisfaction.
- Multi purpose library card.

For Library Users

- Shorter waiting time at circulation desks.
- Higher availability of books due to quick circulation.
- Faster, easier and convenient to use.
- 24 hours book return facility.
- Easy book searching.
- Library cards can be used for transaction in canteen and attendance purpose.
- Better library experience.
- Exposure to new technology.





Feature	RFID Tags	Barcodes
Read more than one item at a time	Yes	No
Read while item is moving	Yes	No
User Programmable	Yes	No
Line of site read not required	Yes	No
Lifetime guarantee	Yes	No
Sturdy in harsh environments	Yes	No
Able to resist water damage	Yes	No
Built-in security	Yes	No
Inventory without handling items	Yes	No
Able to locate specific items on shelves	Yes	No
Use with automatic returns units	Yes	No

Communication Flow, Protocols & Standards

KOHA

LIBSOFT

VTLS

SOUL

LIBERTY

ALICE

LIBSYS

LMS DATABASE

SIP2--NCIP SERVER

SIP2

NCIP

REQ

REPLY

SIP2--NCIP CLIENT

MIDDLEWARE



ISO 15693

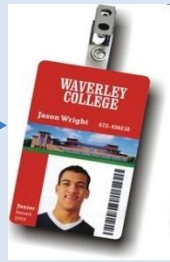
ISO 18000-3



proprietary

ISO 15693

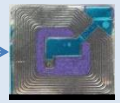
ISO 14443A



Smart card



ISO 15693

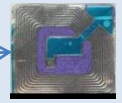


Tag



proprietary

ISO 15693



Tag

Security

EAS

AFI

Virtual DB



RFID LIBRARY COMPONENTS

- Book Tags
- Labels
- Smart Card
- Staff Station
- Handheld Reader
- Book Drop Unit
- Information Kiosk
- Software / Middleware
- Smart Card Printer
- CCTV
- Security Gate
- Automatic Issue Return Kiosk
- Automatic sensor based Sliding gates
- SMS Service pack



RFID Tags

Libraries identification

- Paper thin label.
- Provides Identification and Security in a single tag.
- Pasted on inside of back cover.
- Password protected anti theft securit bit.
- 13.56 MHz frequency.
- ISO 15693, 18000-3
- Maximum read range upto 1 meter in Gate antenna & upto 25 cm in staff station
- Every tag has a unique identification number.
- Memory : up to 2048 bits
- Dimension: Available in various dimensions.
- Data retention: 10 years.









RFID Smart card for Library Users

- Used for book issue and return.
- Mifare multipurpose card. Can be used for attendance, canteen payment, access control, hostel attendance etc
- Frequency: 13.56 MHz
- Memory : 1k.
- Unique ID eliminates the risk of duplicate card.
- Password protected. Can be blocked in case of loss.
- Data retention: 10 years
- Can be used for 10,00,000 transactions.
- Custom printing available.
- Logo, name, address etc of institute can be printed at manufacturing time. (Pre printing)

Automatic Book Issue and Return Kiosk

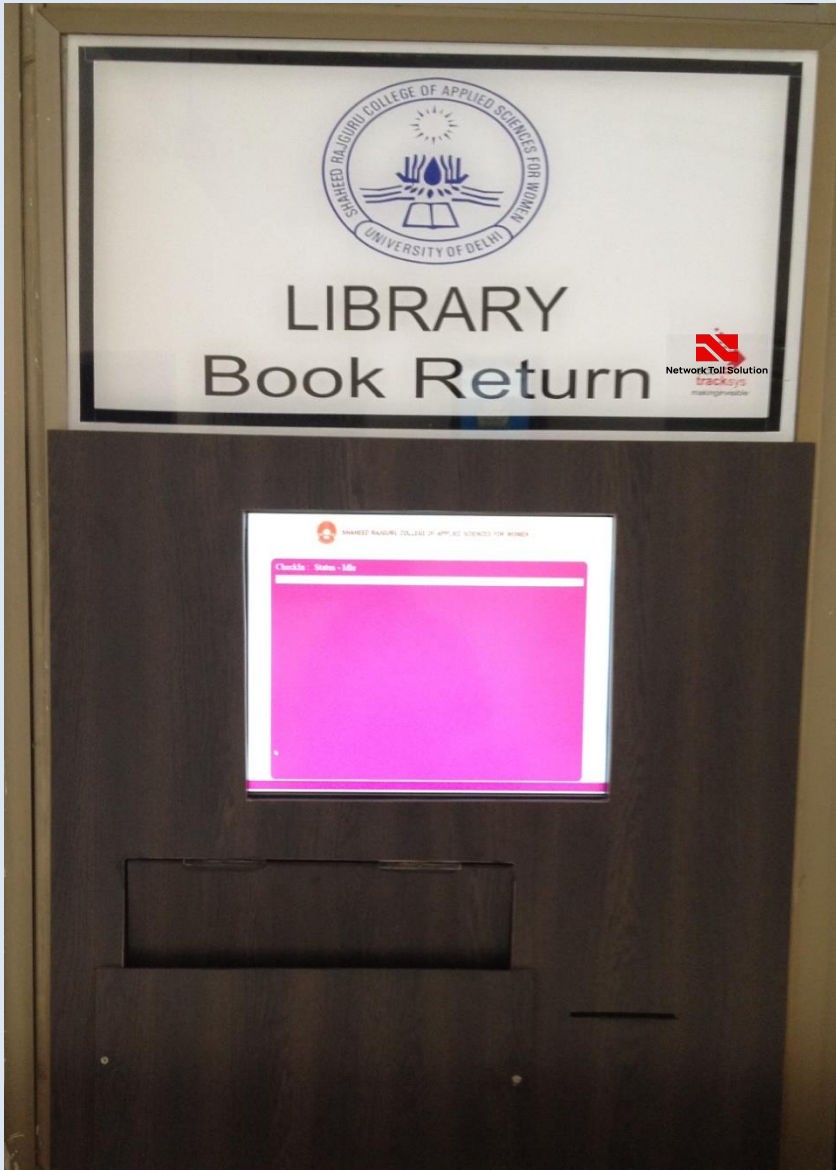




- Used for automatic book issue and return.
- No manual intervention required.
- On every issue and return, print slip with details is issued for future reference.
- 17 inch touch screen kiosk.
- Read Range: Up to 25 cm.
- Self issue and return for 3-4 books at a time.
- Online firmware upgrade.
- ISO 15693 and ISO 14443A compliant.
- NCIP/ SIP 2 Circulation protocol.
- Frequency: 13.56 MHz.
- Antenna size: 300 x 300 mm
- Communication interface: Ethernet.
- Weight: 25 kg approx.
- Esthetic touch.

RFID Book Return Unit





- Installed outside library.
- Wall mounted / independent unit.
- Drop box can be dragged to the shelf with wheels.
- Custom design and colour
- 24*7 book return facility.
- Speeds the book return to library.
- Print slip facility with out of paper notification.
- 17 inch touch screen with wooden esthetic touch.
- Frequency: 13.56 MHz
- NCIP / SIP2 protocol complaint.
- ISO 15693 and 14443A.
- Weight: Approx 20 Kg.
- Communication interface: Ethernet.
- Dimension: 60 x65 x 135 cm

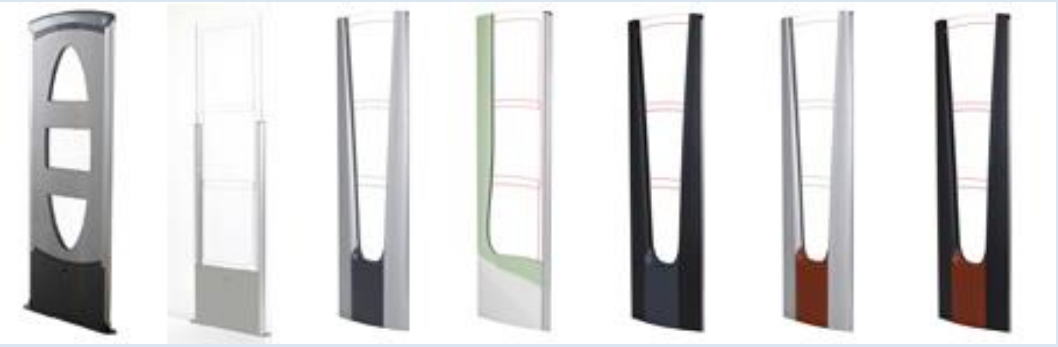


RFID Handheld Reader

- Used for inventory taking and searching a miss located book.
- Read Range: up to 10 cm.
- Window platform.
- internal antenna for stain free operation.
- In built battery.
- Read and write function.
- Wide touch screen display.
- Frequency: 13.5 MHz.
- Power Supply: 5V / 3 A DC Adapter.
- Communication interface: USB
- Storage : 1 GB Extendable.
- Dimension: 19 x 9 x 4.5 cm.
- CPU: 400 MHz.
- Light weighted. 500 gm approx.



RFID Security Gate



- Used for theft prevention.
- Reads in all three orientation.
- Read Range: Up to 1 mt.
- EAS/AFI bit programming inbuilt.
- Integrated buzzer alert system.
- ISO 15693 compliant.
- In built anti collision algorithm.
- API's for customized application.
- Weight : 20 kg max.
- Housing material : ABS
- Dimension: 1530 X 680 X 30 mm
- Communication Interface: RS 232
- Power consumption: 30 V Max.
- Frequency: 13.56 MHz.
- Trans esthetic look.





RFID Staff Station

- Used for book circulation (check in & check out) and registration and renewals.
- Identifies the borrower using smart card.
- Can be used for programming RFID cards.
- Attached to a computer and placed under the table.
- Read Range: 25 cm.
- Multi protocol reader.
- integrated reader and antenna in single unit.
- Plug and play with USB.
- ISO 15693 and 14443A
- 13.56 MHz frequency.
- Power consumption: 500 mw max
- Dimension: 340 x 340 x 20 mm.
- Housing material: Corian Acrylic Polymer.
- Buzzer for read verification.

RFID Information Kiosk for Library

- Used for searching books on entering the library.
- 19 inch touch screen
- Wall mounted / independent unit.
- Weight: Approx 25 kg.
- Custom design and colour to suit the theme.
- In built processor and CPU.
- Space for branding.
- Ventilation unit.
- ISO certified rating.





Smart Card Printer



The most reliable & expandable card printer in its class







Automatic Sliding Gate

- Used at exit point of library.
- Helps in theft prevention.
- Gets activated immediately if gate sounds a buzzer.
- Alternatively can also be used for sensor based door opening and closing for library visitors.
- Weight: 150 kg approx.
- Power : 75 watts.
- Built in 8 nit mIcro computer
- Opening and closing time: 1- 9 sec adjustable.
- Opening adjustable.
- Glass finish.
- Can be integrated with CCTV and security gate.

Thank You!







sales@networktoll.com www.networktoll.com Contact us: +91 9289757018